

ECE 3337: Signals & Systems Analysis

Lecture 18: Fourier Analysis Techniques

**Reference: Chapter 5
(5.2 – 5.3)**

Please turn off your gadgets & other distractions now



Reminder: Exam 2

Syllabus: Chapters 1 - 4, as covered in class

Time: Saturday, April 4th , 10 am – 11:30 am

Location: Science & Research Building room 117 (both sections)

Help Sessions

March 27th 10am – noon, in-person @ Cemo Hall Room 109 (Dr. Roysam)

April 3rd noon – 2pm, Online via MS-TEAMS (Dr. Reddy)

Old Exams with solutions are on the Canvas site for practice

- 1. Try the old exams individually without looking at the answers**
- 2. Attempt the problems and practice until you can do them fully on your own without notes, quickly, just like in a test.**
- 3. Practice for correctness and speed**



Recap: Cosine-reference Phasors

Phasors are a concise way of describing signals at a fixed and known frequency

Given a “forever” signal:

$$v(t) = V_0 \cos(\omega t + \phi)$$

Its phasor representation is:

$$\mathbf{V} = V_0 e^{j\phi}$$

$$y(t) \rightarrow \mathbf{Y}$$

$$\frac{dy}{dt} \rightarrow j\omega \mathbf{Y}$$

$$\frac{d^2 y}{dt^2} \rightarrow (j\omega)^2 \mathbf{Y}$$

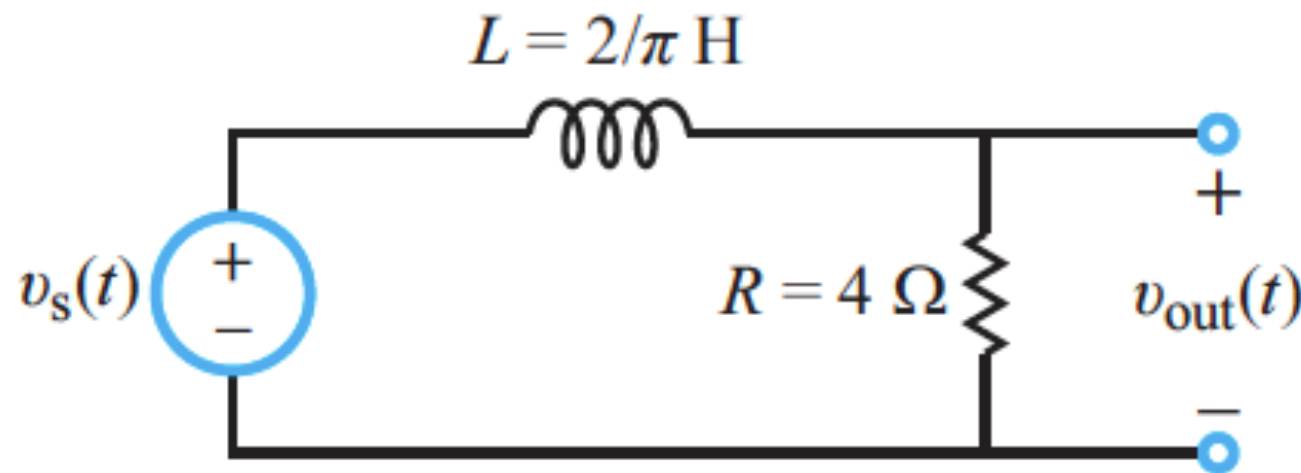
Note: The frequency ω is not explicitly written

– it is implicit i.e., understood so not stated explicitly



Circuit Analysis with Phasors

- We use the same method as with the Laplace transform
- Just replace s with $j\omega$, and no initial conditions
- Example:



$$\mathbf{V}_{out} = \left(\frac{R}{R + j\omega L} \right) \mathbf{V}_s \quad \text{VOLTAGE DIVIDER}$$



Jean-Baptiste Joseph Fourier

**Famous French Mathematician
(1768 – 1830)**

- **Best known for inventing the Fourier Series**
- **Made fundamental contributions to the study of vibrations and heat transfer**
 - **Discovered the greenhouse effect**





Fourier's Grand Claim

Any general periodic signal

$$x(t) = x(t + nT_0)$$

can be expressed as a linear combination of sinusoids, as follows:

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

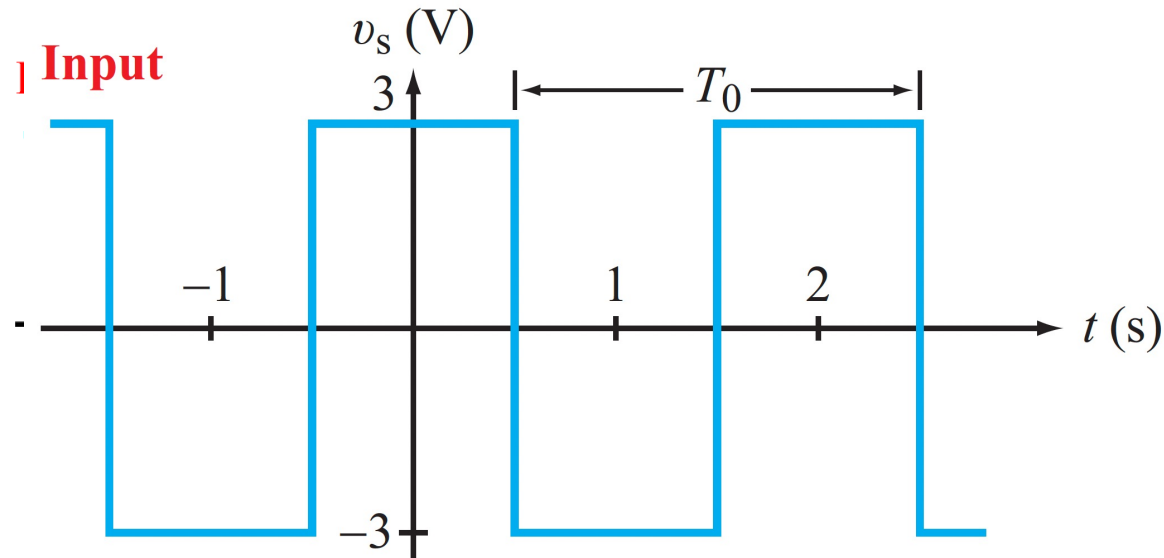
$$= \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$$

WHY IS THIS A BIG DEAL?



Example

Consider the following square waveform



FOURIER SERIES REPRESENTATION (ACCEPT THIS FOR NOW)

$$v_s(t) = \frac{12}{\pi} \left(\cos \omega_0 t - \frac{1}{3} \cos 3\omega_0 t + \frac{1}{5} \cos 5\omega_0 t - \dots \right)$$

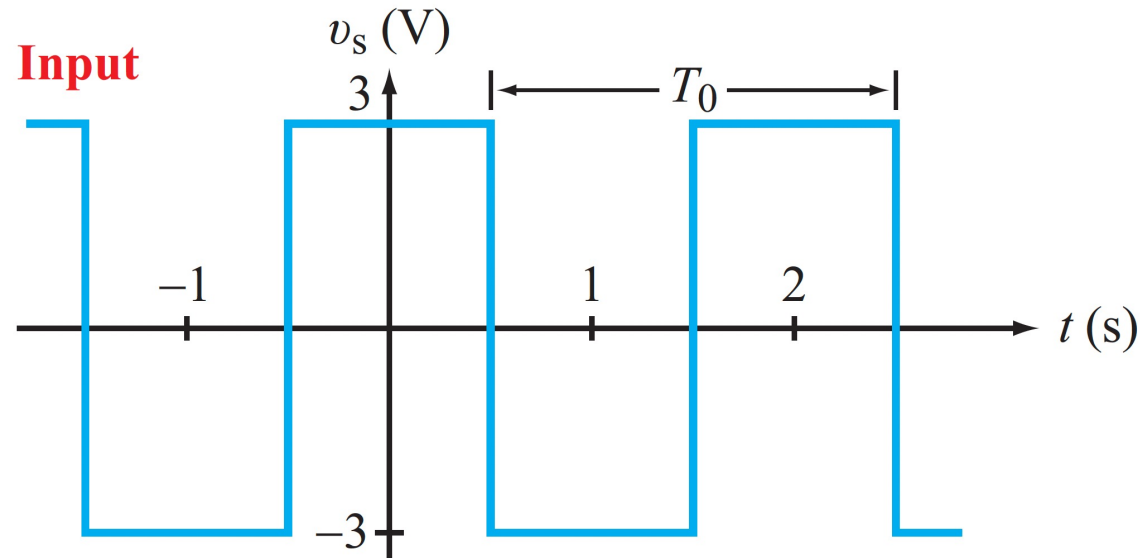
**WE WILL
ACTUALLY DERIVE
THIS SOON**

$$\omega_0 = \frac{2\pi}{T_0} \quad \text{FUNDAMENTAL ANGULAR FREQUENCY}$$



Example

Consider the following square waveform



FOURIER SERIES REPRESENTATION

$$v_s(t) = \frac{12}{\pi} \left(\cos \omega_0 t - \frac{1}{3} \cos 3\omega_0 t + \frac{1}{5} \cos 5\omega_0 t - \dots \right)$$

WE WILL
ACTUALLY DERIVE
THIS SOON

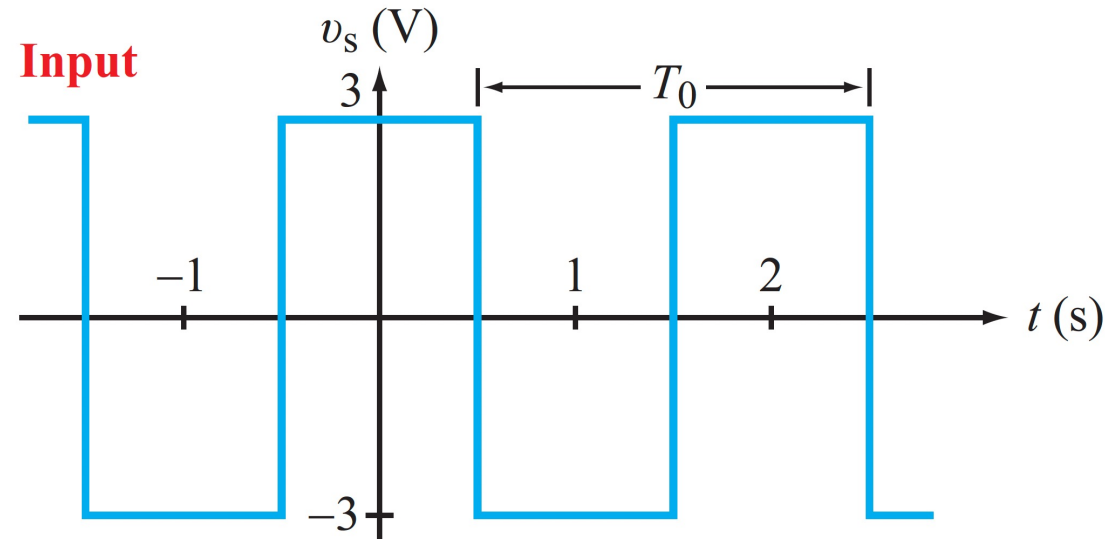
$$m\omega_0 = \frac{2\pi m}{T_0}$$

HARMONIC FREQUENCIES



Example

Consider the following square waveform



FOURIER SERIES REPRESENTATION

$$v_s(t) = \frac{12}{\pi} \cos \omega_0 t - \frac{12}{3\pi} \cos 3\omega_0 t + \frac{12}{5\pi} \cos 5\omega_0 t - \dots$$

WE WILL
ACTUALLY DERIVE
THIS NEXT CLASS

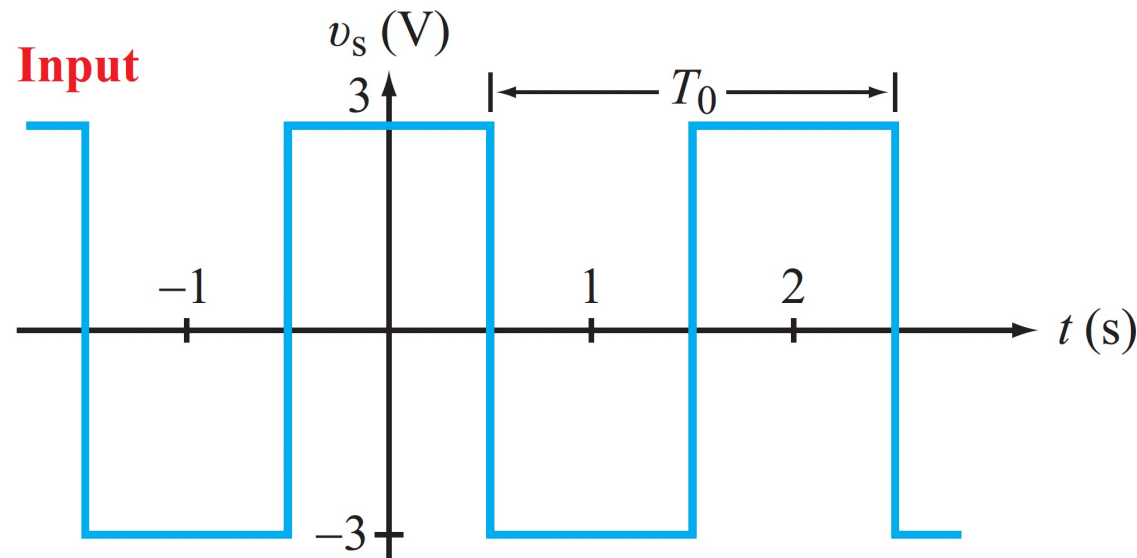
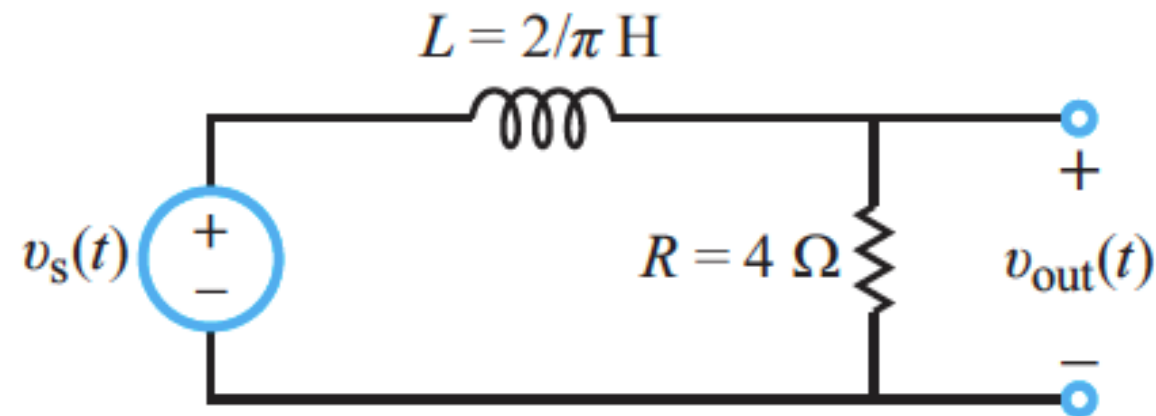
$$m\omega_0 = \frac{2\pi m}{T_0}$$

THIS EXAMPLE ONLY CONTAINS ODD
HARMONICS!
IT IS AN EVEN FUNCTION



Motivating Example

Consider the following L-R circuit that is excited by a periodic square waveform



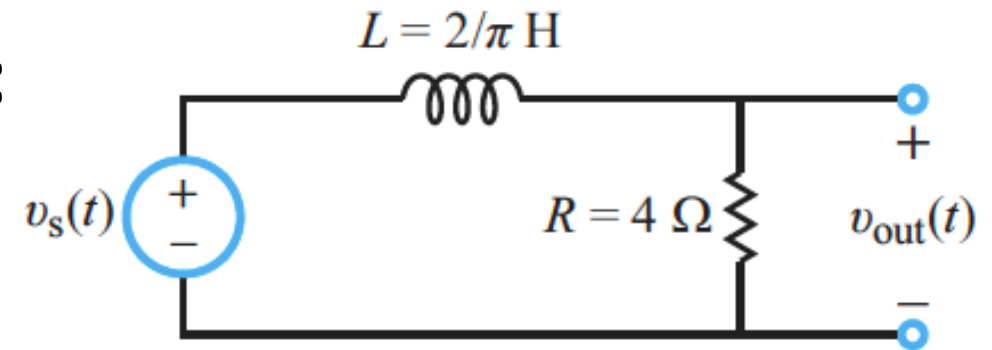
$$\omega_0 = \frac{2\pi}{T_0} = \pi$$



Strategy

- Keep in mind that a phasor representation is for a **FIXED FREQUENCY** that is not explicitly stated
- However, we can do phasor analysis one harmonic at a time, treating each harmonic as a fixed frequency
- By the voltage divider rule, we have:

$$\mathbf{V}_{out} = \left(\frac{R}{R + j\omega L} \right) \mathbf{V}_s$$

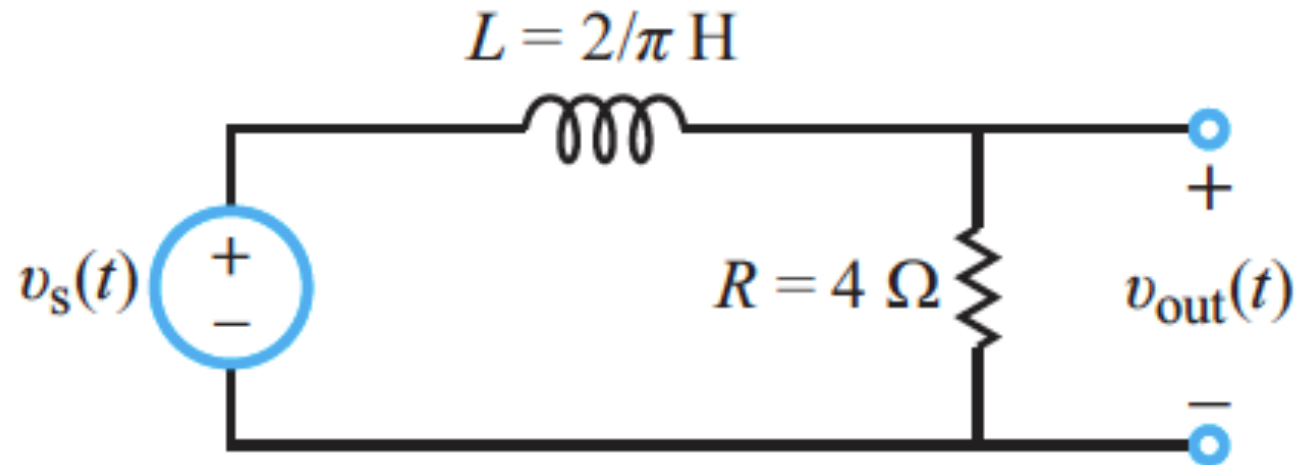


- We can do phasor analysis separately at each harmonic:

$$\omega = \omega_0, 3\omega_0, 5\omega_0 \dots$$



Circuit Responses by Frequency



$$\mathbf{V}_{\text{out}1} = \left(\frac{R}{R + j\omega_0 L} \right) \mathbf{V}_{s1}$$

$$\mathbf{V}_{s1} = \frac{12}{\pi}$$

$$\mathbf{V}_{\text{out}3} = \left(\frac{R}{R + j3\omega_0 L} \right) \mathbf{V}_{s3}$$

$$\mathbf{V}_{s3} = -\frac{12}{3\pi}$$

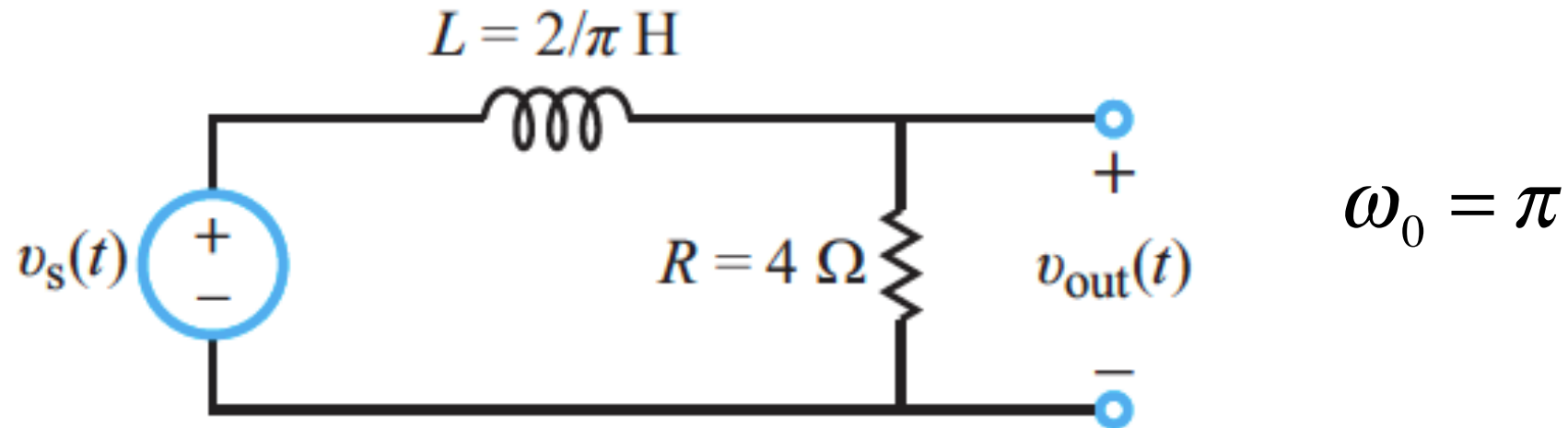
$$\mathbf{V}_{\text{out}5} = \left(\frac{R}{R + j5\omega_0 L} \right) \mathbf{V}_{s5}$$

$$\mathbf{V}_{s5} = \frac{12}{5\pi}$$

• • •



Circuit Responses by Frequency



$$\mathbf{V}_{out1} = \left(\frac{4}{4 + j\pi 2/\pi} \right) \mathbf{V}_{s1} = 3.42e^{-j26.56^\circ}$$

$$\mathbf{V}_{out3} = \left(\frac{4}{4 + j3\pi 2/\pi} \right) \mathbf{V}_{s3} = -0.71e^{-j56.31^\circ}$$

$$\mathbf{V}_{out5} = \left(\frac{4}{4 + j5\pi 2/\pi} \right) \mathbf{V}_{s5} = 0.28e^{-j68.2^\circ}$$

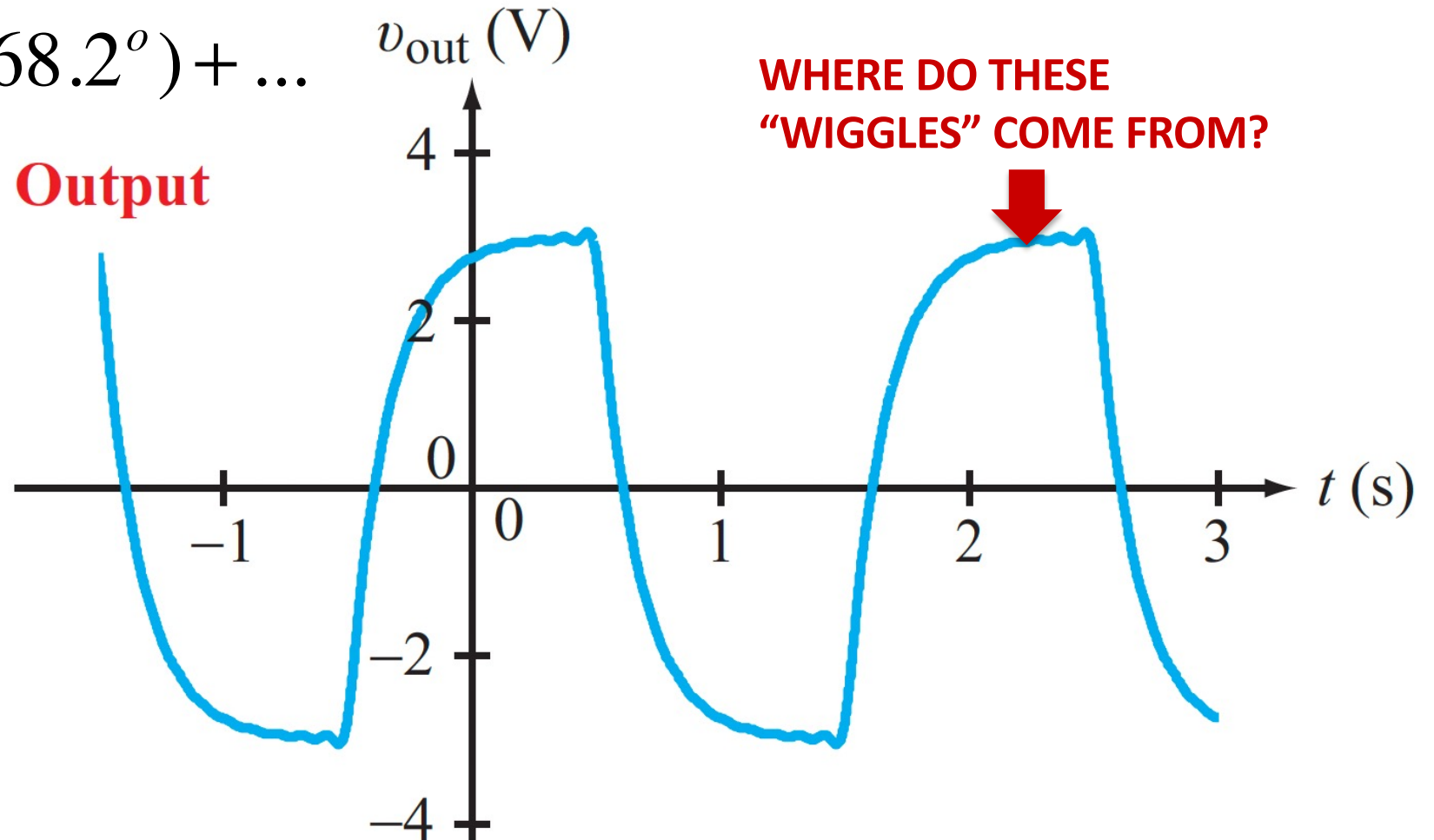


Apply Superposition to Get the Overall Response

$$v_{out}(t) = 3.42 \cos(\omega_0 t - 26.56^\circ) \\ - 0.71 \cos(3\omega_0 t - 56.31^\circ) \\ + 0.28 \cos(5\omega_0 t - 68.2^\circ) + \dots$$

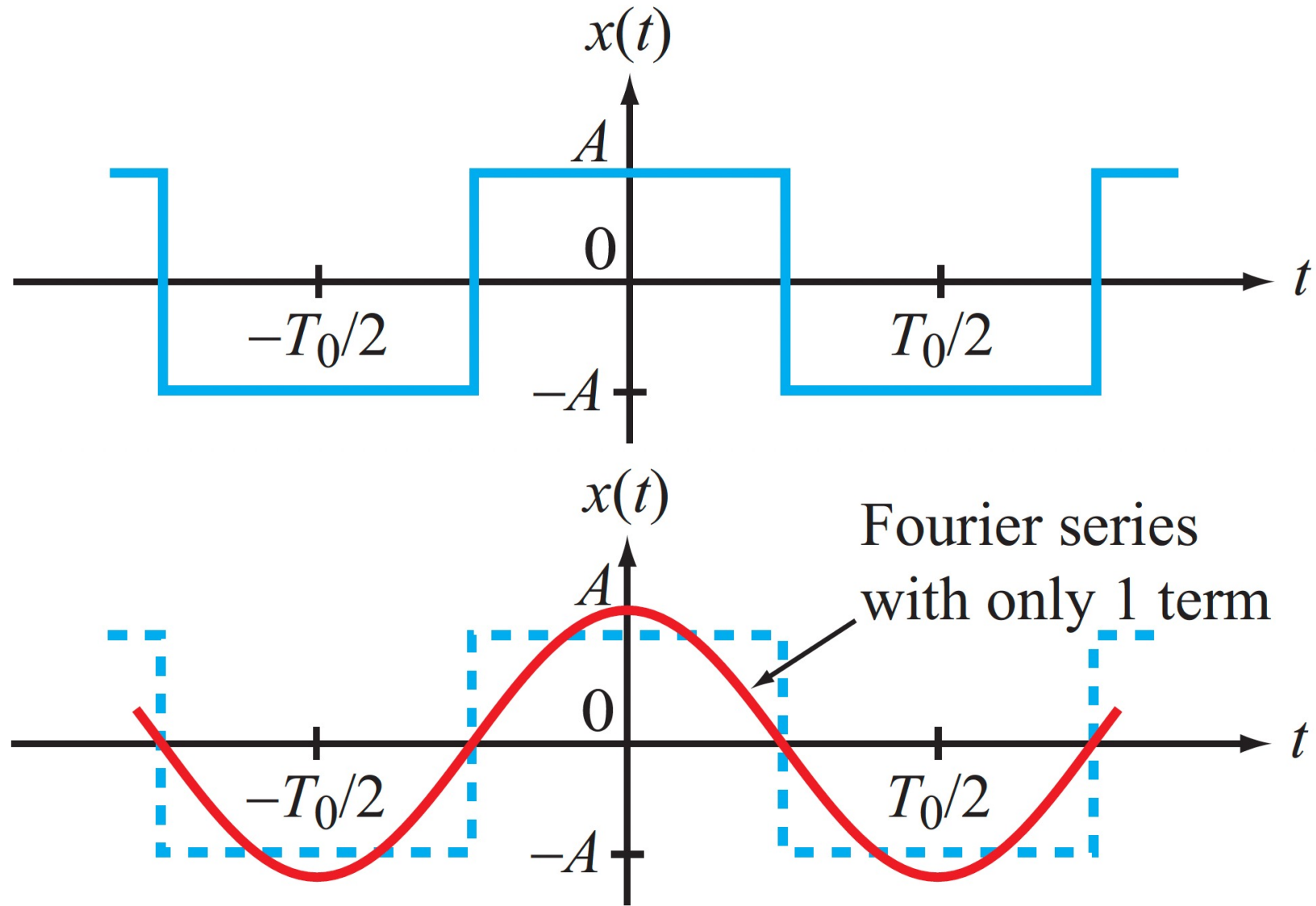
**AMAZINGLY GOOD
APPROXIMATION !!**

**GETS EVEN BETTER IF
WE USE MORE TERMS**



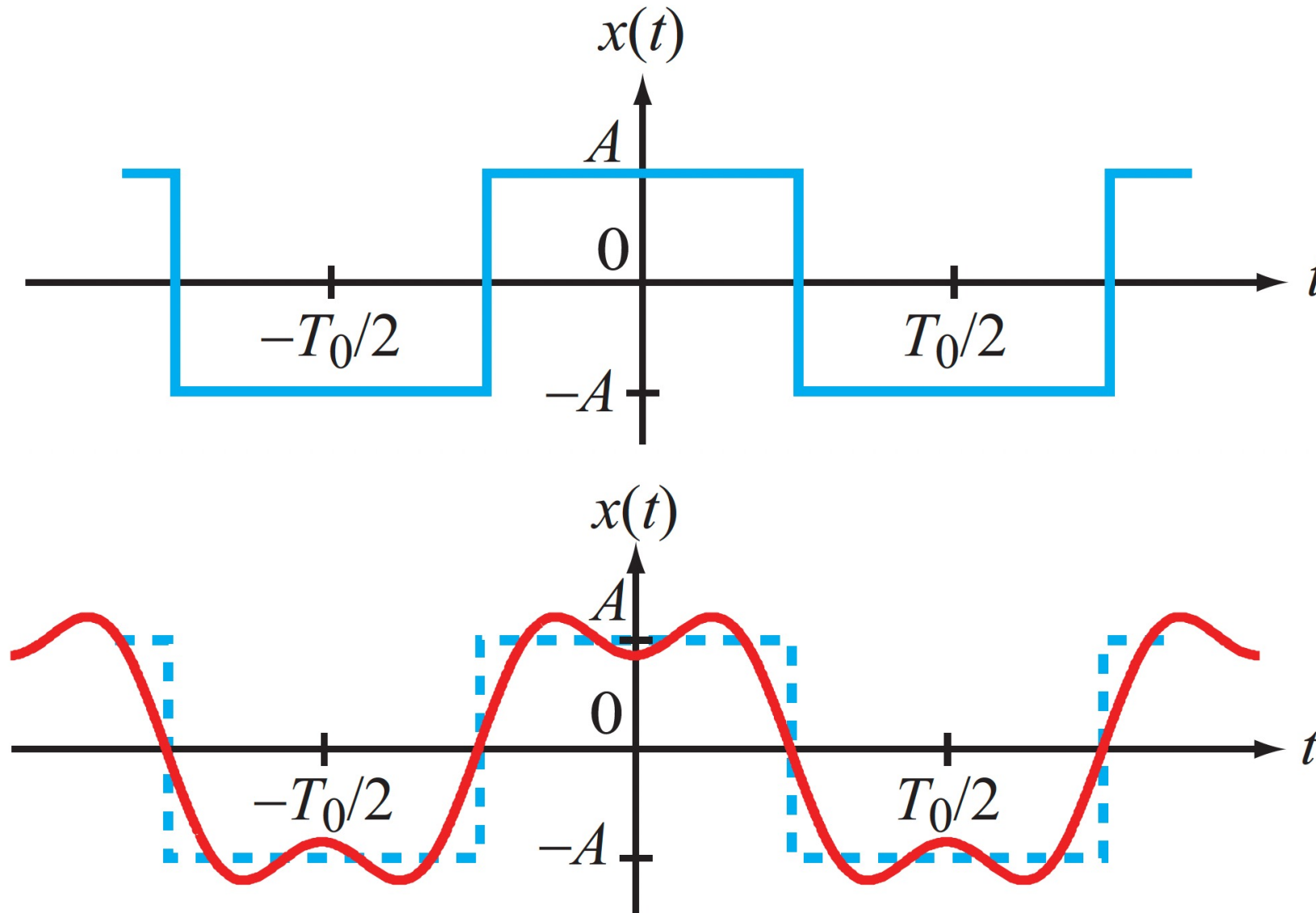


The First Term



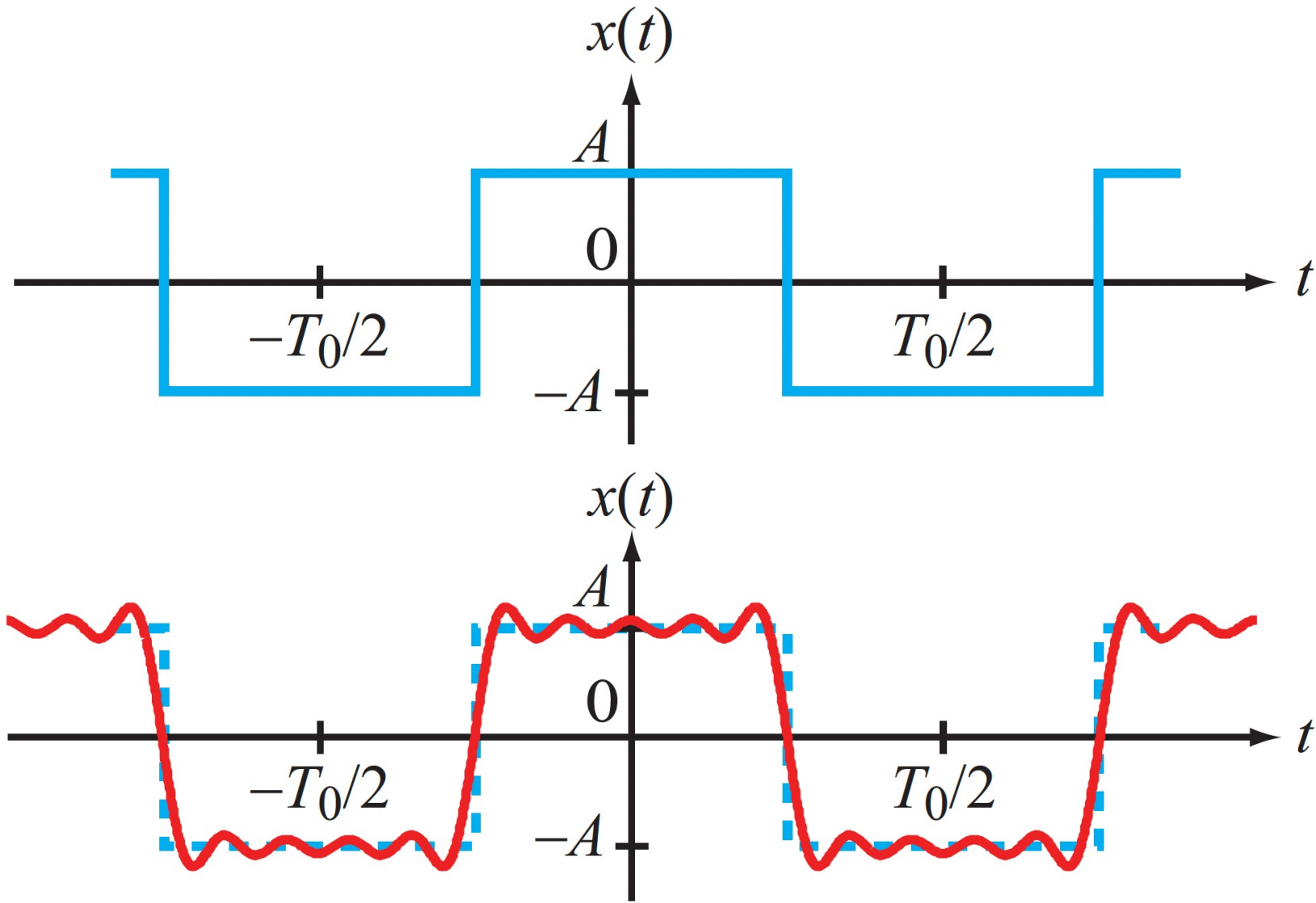


The First 3 Terms



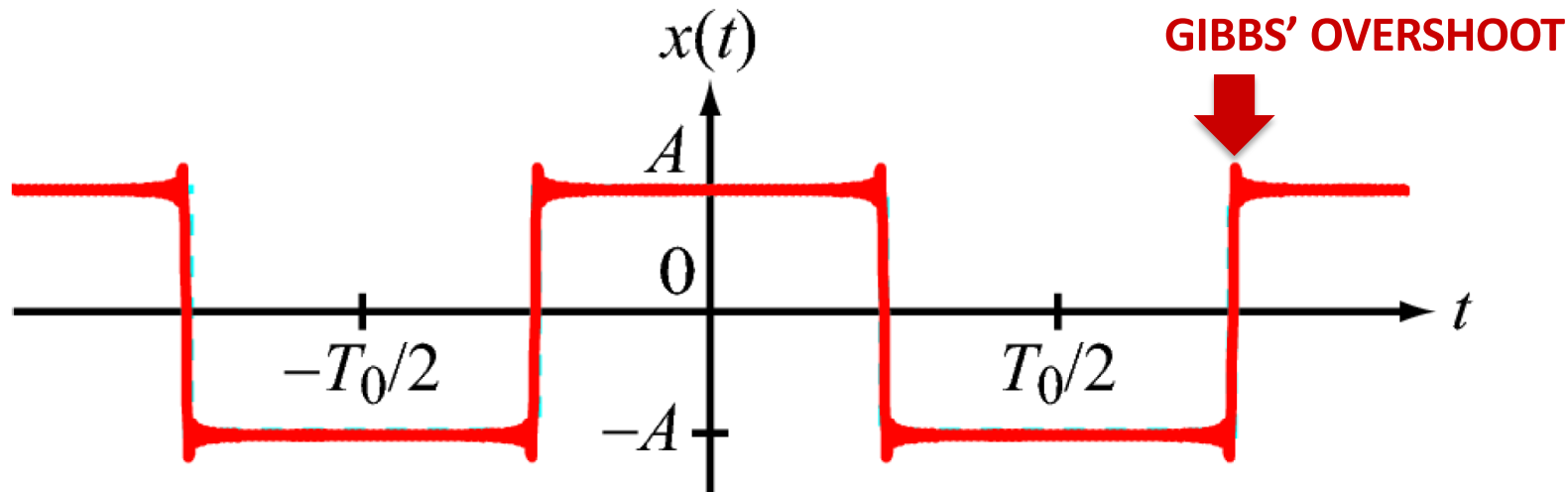
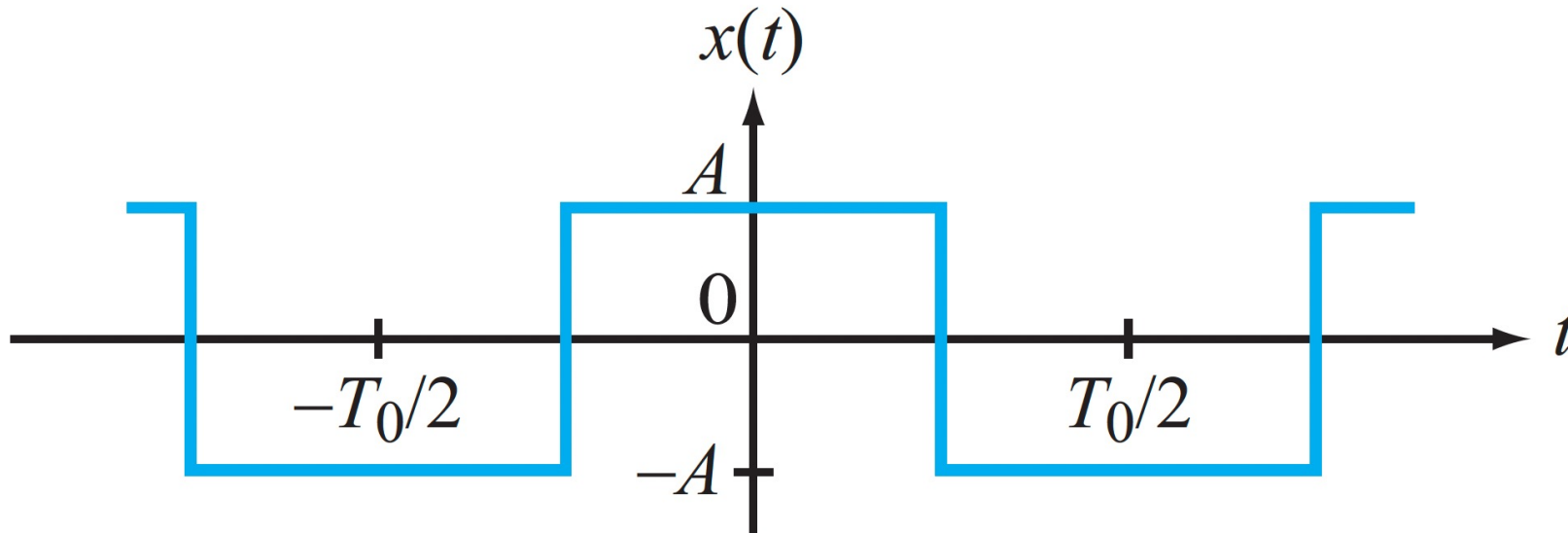


The First 10 Terms

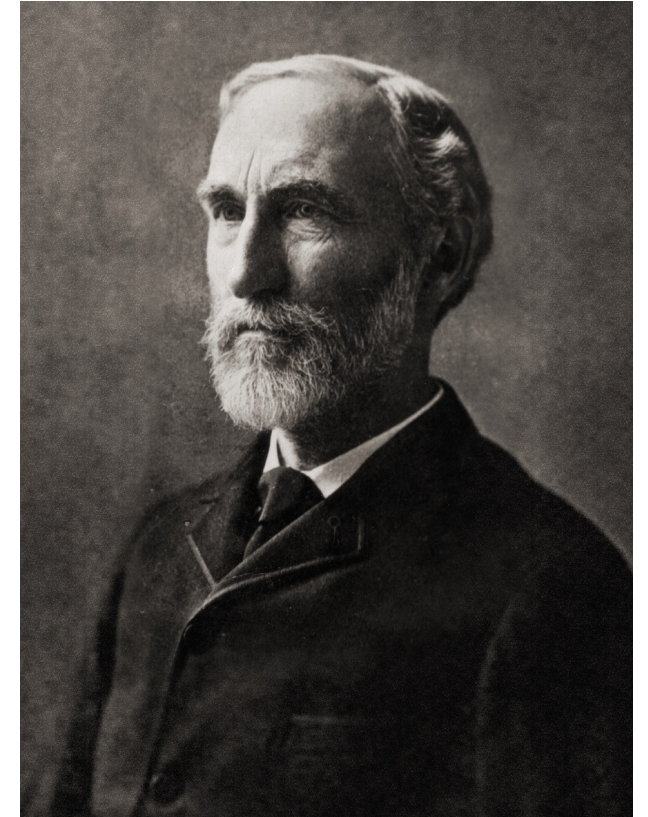




The First 100 Terms



(e) Fourier series with 100 terms



Josiah Willard Gibbs
(America's First PhD, 1899)



Trigonometric Identities

Euler's formula can be used to derive common trigonometric formulas

$$e^{j(A+B)} = \cos(A+B) + j \sin(A+B)$$

$$\begin{aligned} e^{jA} e^{jB} &= (\cos A + j \sin A) \times (\cos B + j \sin B) \\ &= \cos A \cos B - \sin A \sin B + j(\sin A \cos B + \cos A \sin B) \end{aligned}$$

COMPARING THE CORRESPONDING TERMS, WE GET:

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$



Derivation of Fourier Series

A general periodic signal

$$x(t) = x(t + nT_0)$$

can be expressed as a linear combination of sinusoids, as follows:

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$

LET'S TRY TO DERIVE THIS FORMULA ...



Trigonometric Identities

Similarly,

$$e^{j(A-B)} = \cos(A-B) + j \sin(A-B)$$

$$\begin{aligned} e^{jA} e^{-jB} &= (\cos A + j \sin A) \times (\cos B - j \sin B) \\ &= (\cos A \cos B + \sin A \sin B) + j(\sin A \cos B - \cos A \sin B) \end{aligned}$$

COMPARING THE CORRESPONDING REAL & IMAGINARY TERMS, WE GET:

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$



Additional Identities

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y \text{ (basic formula 1)}$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y \text{ (basic formula 2)}$$

$$\Rightarrow \sin(2x) = 2 \sin x \cos x$$

$$\Rightarrow \cos(2x) = \cos^2 x - \sin^2 x$$

$$\Rightarrow \sin x \sin y = \frac{1}{2} [\cos(x - y) - \cos(x + y)]$$

$$\Rightarrow \cos x \cos y = \frac{1}{2} [\cos(x - y) + \cos(x + y)]$$

ETC.

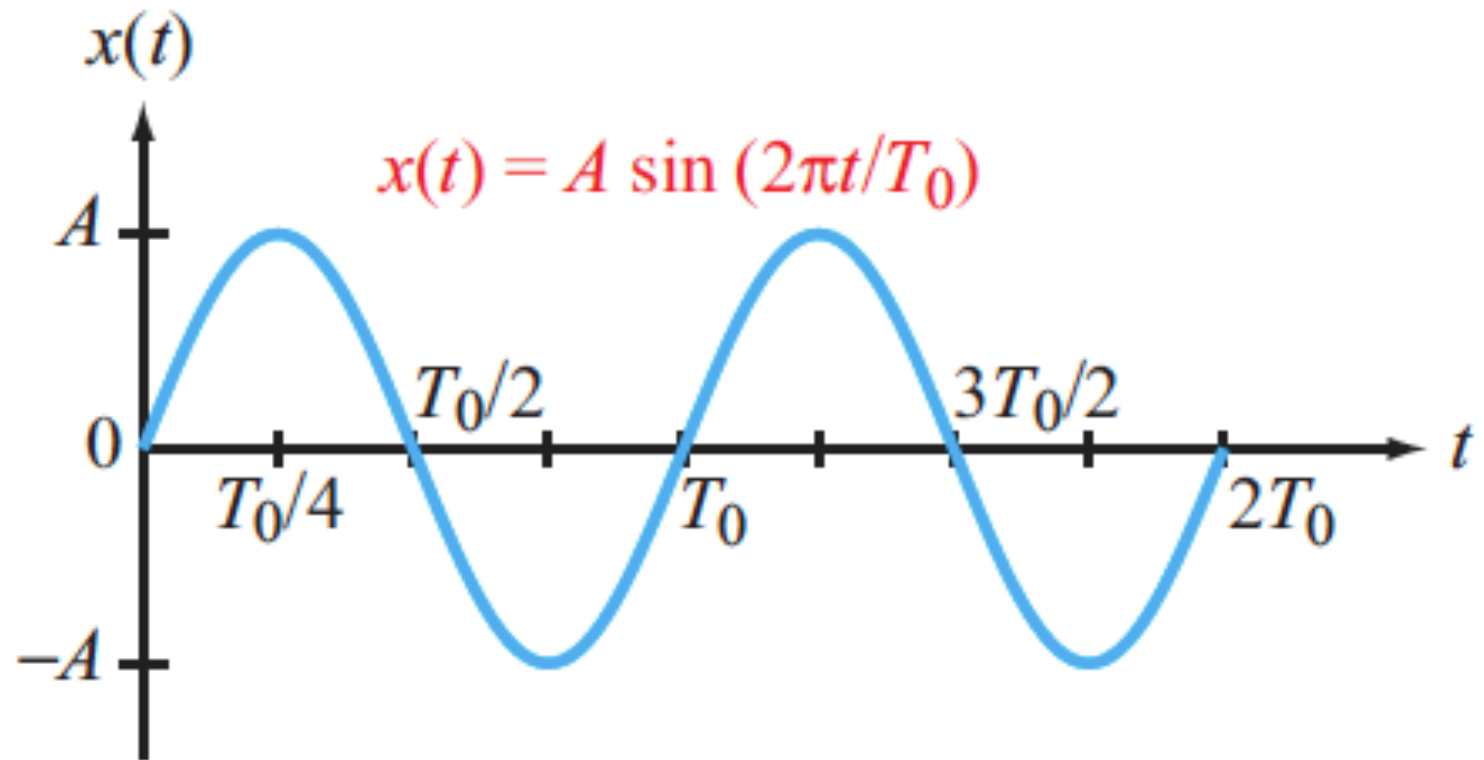


Representing Phase Shifts & Delays

$$\begin{aligned}x(t) &= A \sin\left(\frac{2\pi t}{T_0} + \theta\right) \\&= A \sin(\omega_0 t + \theta) \\&= A \sin(\omega_0 t + \omega_0 \tau)\end{aligned}$$

$$\omega_0 = \frac{2\pi}{T_0} = 2\pi f_0$$

τ = Time delay equivalent to angle θ



$$\text{When } \tau = \frac{T_0}{4}, \quad \theta = \frac{\pi}{2} = 90^\circ$$



Some Terminology

A Fourier series is written in terms of a fundamental frequency, and its multiples, called harmonics

$$\omega_0 = \frac{2\pi}{T_0} = \text{Fundamental Angular Frequency}$$

$$f_0 = \frac{1}{T_0} = \text{Fundamental Frequency}$$

$$n \times \omega_0 = n^{\text{th}} \text{ Harmonic}$$

$$\{a_n, b_n\} = \text{Fourier Coefficients}$$



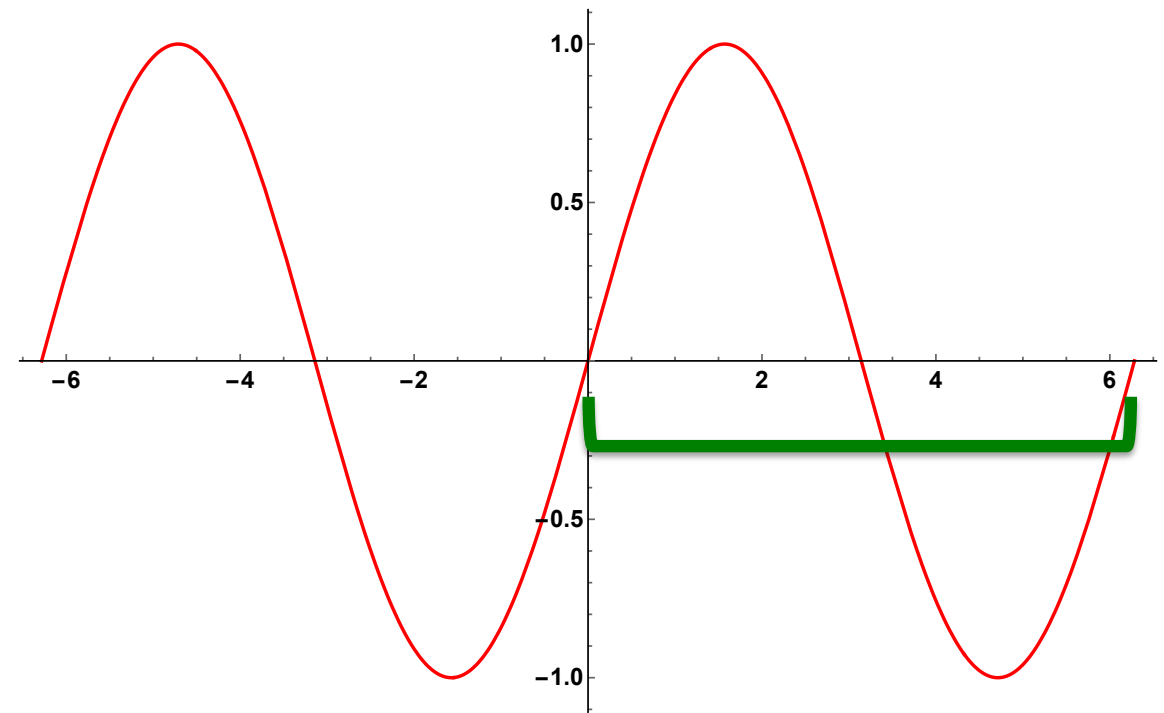
Integrals of Sinusoids

The integral of a sinusoid over one time period is zero

$$\int_0^{T_0} \sin(\omega_0 t) dt = 0$$

By extension, this is also true over multiple periods:

$$\int_0^{T_0} \sin(n\omega_0 t) dt = 0$$





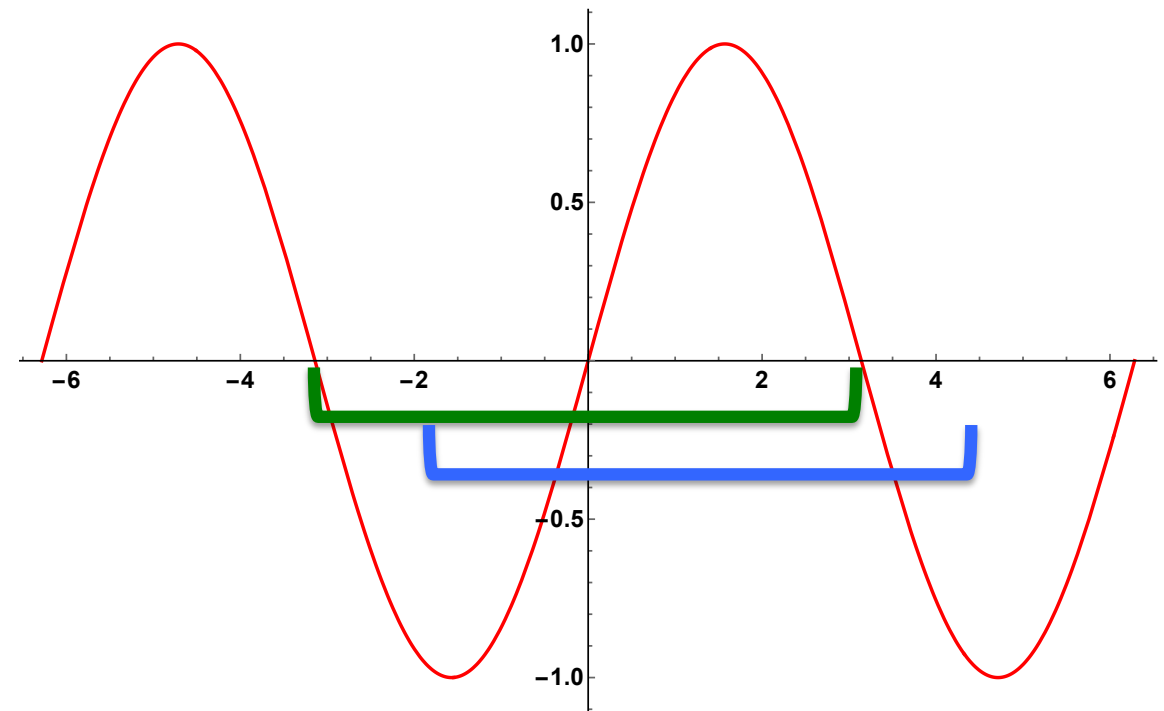
Integrals of Sinusoids

Introducing a phase shift does not change the answer, since it merely re-arranges the same set of points

$$\int_0^{T_0} \sin(n\omega_0 t + \phi) dt = 0$$

Similarly,

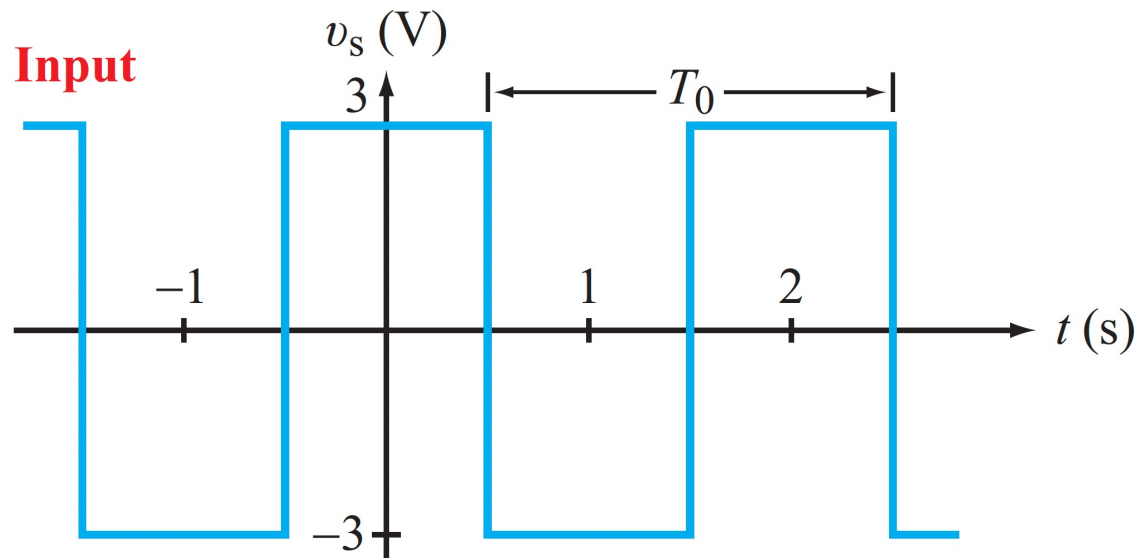
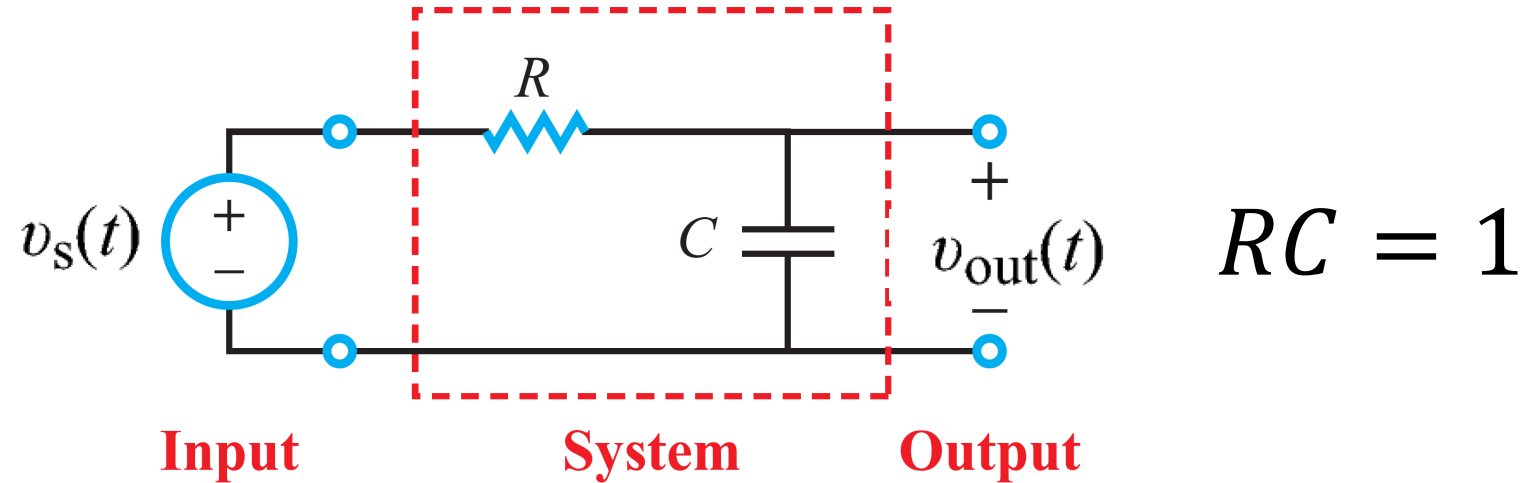
$$\int_0^{T_0} \cos(n\omega_0 t + \phi) dt = 0$$





Activity

Calculate the first 3 terms of the output, at $(\omega_0, 3\omega_0, 5\omega_0)$ for the RC circuit:



$$\omega_0 = \frac{2\pi}{T_0} = \pi$$



Summary

We learned two important and related concepts today:

1. Phasors

2. Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

- **Complicated periodic signals can be approximated quite well by calculating the Fourier series to a limited number of terms**
 - **We can use phasor analysis to calculate the LTI response to each sinusoid (harmonic), one at a time**
 - **Add the sinusoid responses to calculate the overall response**
 - **Use more terms to get a better approximation**



For the Next Class

- **Work through the textbook examples in Section 5-1 – 5-2**
- **Study Sections 5-3 – 5.4.2**
- *Try demos at <https://betterexplained.com/articles/an-interactive-guide-to-the-fourier-transform/>*

How to Study:

1. **Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**